

Natural History Museum (Kids)

UK Family Tour

Hope: The Biggest Animal That Ever Lived
Look up. The skeleton diving toward you through Hintze Hall is Hope, a blue whale — twenty-five meters long, four and a half tonnes of bone. Blue whales are the largest animals that have EVER existed. Bigger than any dinosaur. A blue whale's heart is the size of a small car, and a child could crawl through its biggest blood vessels. Its tongue alone weighs as much as an elephant. And yet it eats some of the smallest creatures in the ocean — krill, little shrimp — about four tonnes of them a day, filtered through bristly plates in its mouth. Hope is a real whale that stranded in Ireland in 1891. She was in the museum's storage and side galleries for over a century before taking the star spot in 2017, replacing Dippy the Diplodocus — which caused a national argument, because Britain loved that dinosaur. But there's a reason she's called Hope: humans hunted blue whales almost to extinction — down to a few thousand left — and then the world agreed to stop, and the whales are slowly coming back. She hangs in a diving position, like she's lunging for krill, as a reminder that we nearly lost the biggest animal in history, and chose not to.

How to Hang a Whale

Here is a puzzle. How do you hang a skeleton the size of an aeroplane from a ceiling, so that it looks like it is diving, and so that it never, ever falls on anybody? That was the job the museum gave a team of engineers and bone experts, and it took them three whole years just to plan.

First they had a problem you might not guess. Whale bones are full of oil. A blue whale's skeleton keeps leaking oil for decades after the animal has died, slowly, like a very slow candle. So before Hope could go on show, conservators had to clean and de-grease every single bone, or she would have dripped on the visitors below. Then each bone was scanned by laser to make a perfect three-dimensional map, so the team could work out exactly how the whole thing balanced.

Before all that, Hope had a long and rather sleepy history. For years she hung in an older gallery, high above a giant model of a blue whale that workers built out of wood and

plaster back in the nineteen thirties. And here is the museum's favourite legend. The men who built that model left a little trapdoor in the whale's belly, and the story goes that they used to climb inside and eat their lunch in there. People still whisper that when they finally sealed it up, the builders hid a few coins, a telephone book, and maybe even a bottle inside the belly of the fake whale. Nobody is completely sure. It has never been opened.

When the day came to move the real Hope into the grand entrance hall, the team lifted her bone by bone on cranes and fixed her to a hidden metal frame, curved into a lunging pose, as if she has just this second spotted a cloud of tiny shrimp and is twisting to swallow them. Look at her spine. See how it bends? A dead skeleton cannot bend itself. Every one of her bones is held at exactly the right angle by that clever frame, so that several tonnes of whale can float over your head and seem completely alive.

So when you stand under her, remember, you are not just looking at a dead animal. You are looking at three years of work, a laser map of every bone, and possibly a secret lunch spot sealed inside a whale forever.

A Day in the Life of a Blue Whale

Imagine you could shrink down and follow a living blue whale for one whole day. What would it be like? First, you would need good ears, because a blue whale is one of the loudest animals on the entire planet. Its call can reach a hundred and eighty-eight decibels, which is louder than a jet plane taking off. But you would barely hear it, because most of that sound is so low and deep that human ears cannot pick it up. Other whales can. A blue whale's rumble travels through the ocean for hundreds of miles, which means two whales on opposite sides of a whole sea can slowly call back and forth to each other, like the world's biggest, slowest telephone.

Now hold your breath, because we are going down. A feeding blue whale dives deep and can hold its breath for many minutes at a time, its enormous heart slowing right down to save oxygen. Down in the dark it hunts clouds of krill, tiny shrimp smaller than your finger. And here is

the amazing part. The whale does not nibble. It speeds up, opens its mouth impossibly wide, and gulps a swimming pool of water and krill all at once. Its throat has deep pleats like an accordion that balloon out to hold it all. Then it squeezes the water back out through bristly plates and swallows what is left. One giant gulp can hold more water than the whale's whole body weighs.

Now, how long does our whale live? A blue whale can live to eighty or ninety years old, about the same as a person, and a few may reach a hundred and ten. Think about that for a moment. Somewhere in the ocean right now there may be an old blue whale that was born before humans agreed to stop hunting them, an animal that has lived through that entire rescue.

And blue whales begin life in a tremendous hurry. A baby blue whale is the fastest-growing animal on Earth. It drinks hundreds of litres of its mother's thick milk every day and puts on weight faster than almost anything alive, growing by roughly the weight of a grown person each and every day in its first months. From tiny shrimp, the biggest animal that has ever lived just keeps getting bigger. Not a bad day's work.

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Dinosaurs: The Word Was Invented Here

Here's something amazing: the word "dinosaur" was invented by the man who built this museum. In 1842, scientist Richard Owen looked at giant fossil bones and realized they belonged to a whole lost group of animals. He named them Dinosauria — "terrible lizards." Then he spent decades fighting for a "cathedral to nature" grand enough to hold them: this building, opened in 1881. In the dinosaur gallery, look for three things. First, the roaring animatronic T. rex — younger kids sometimes bail, so decide your bravery level in advance. Second, the first *Iguanodon* fossils ever studied: early scientists found a spike and stuck it on the nose, like a rhino horn. Totally wrong — it was a thumb spike! *Iguanodon* basically had built-in daggers for thumbs. Scientists make mistakes and fix them; that's the whole job. Third, look for fossils collected by Mary Anning, a poor girl from the seaside town of Lyme Regis who, starting when she was about twelve, in the early 1800s, dug up some of the first ichthyosaurs and plesiosaurs — sea monsters of the dinosaur age. The fancy scientists of London bought her fossils and often took the credit. Today she's celebrated as one of the greatest fossil hunters ever. Not bad for a twelve-year-old.

Dippy the Dinosaur's Big Adventure

Before Hope the whale hung in the great hall, somebody else stood there for a very long time. His name was Dippy, and he was a *Diplodocus*, a long-necked dinosaur so enormous that his neck and tail together stretched more than twenty-six metres, nose to tail-tip, right across the hall.

But here is a secret about Dippy. He is not made of real bones. Dippy is a copy, a cast made of plaster, painted to look like fossil. The real *Diplodocus* was dug up in America, and a very rich man named Andrew Carnegie was so proud of it that he had copies made and sent them to museums all over the world. London's Dippy arrived more than a hundred years ago, in nineteen hundred and five. For decades children grew up meeting Dippy, and from the nineteen seventies he stood right in the centre of the entrance hall, the first thing you saw when you walked in.

So when the museum announced in twenty seventeen that Dippy would be taken down and a whale would take his place, a lot of people were upset. There were newspaper headlines. There was even a campaign called Save Dippy. Britain loved that dinosaur.

But Dippy did not retire. He went on the adventure of a lifetime. Because he is a cast, not fragile fossil, he could travel, so the museum took him apart, packed him into crates, and sent him on a tour all around the United Kingdom. Dippy stood in cathedrals with stained glass windows soaring above him. He visited museums in cities far from London, in Scotland, in Wales, in Northern Ireland. Millions of people who could never get to London came to meet him, many of them seeing a dinosaur up close for the very first time.

And where is Dippy now? After his grand tour he settled into a museum in the city of Coventry, where you can still go and stand under him today. So both stories have a happy ending. Hope the whale reminds us to protect the living world, and Dippy the dinosaur went out and travelled the whole country, meeting more people than he ever met standing still in London. Two great giants, and the museum found a clever way to share them both.

The Dinner Inside a Dinosaur

Here is one of the strangest true stories in the whole history of dinosaurs. On the last night of the year eighteen fifty-three, a group of important gentlemen in top hats sat down to a fancy dinner, with soup and roast and singing. And they ate it sitting inside a dinosaur.

Let me explain. Long before this museum was built, people had only just realised that dinosaurs had ever existed at

all. So a sculptor named Benjamin Waterhouse Hawkins was given an amazing job. He was asked to build the very first life-sized dinosaur models the world had ever seen, out of brick, iron, and concrete, for a great glass building in south London called the Crystal Palace. The scientist Richard Owen, the same man who later helped create this museum, told him what the animals should look like.

To make the giant Iguanodon, Hawkins first built a hollow mould, a bit like an enormous empty shell. And before he filled it, he had a wonderful idea. He set a table for about twenty guests inside the mould of the Iguanodon and threw a New Year's Eve party in there. Owen sat at the head of the table, right where the dinosaur's head would be. They ate a feast of many courses and sang songs late into the night, tucked inside a concrete dinosaur.

Now here is the funny part. Owen and Hawkins got a lot of it wrong. Nobody yet understood these animals properly, so they built the Iguanodon as a big lumpy creature on four legs, like a giant lizard, with a pointed spike stuck on its nose like a rhinoceros horn. Years later scientists worked out that the spike was not a nose horn at all. It was a thumb. And Iguanodon did not lump along on all fours the way they thought.

So the Crystal Palace dinosaurs are, in a way, a giant beautiful mistake. But people love them anyway. They are still standing in a park in London today, more than a hundred and seventy years later, so old and so important that they are now specially protected, like a precious building. You can go and visit them, these funny, wrong, wonderful concrete monsters, the great-great-grandparents of every dinosaur model you have ever seen. And you can stand right beside the very Iguanodon that once had a dinner party inside its belly.

Meet Sophie and the Real T. rex

The museum has a roaring, moving T. rex that lunges and growls, and it is brilliantly scary. But it is a robot. So let me tell you what the real animal was actually like, and then introduce you to a real dinosaur skeleton you can meet here too.

Start with *Tyrannosaurus rex*. It was one of the biggest meat-eating animals ever to walk on land, up to about thirteen metres long, as long as a bus and a bit more. Its head alone was longer than you are tall, and its mouth was full of teeth the size of bananas, some as long as your forearm, with rough edges like steak knives. Scientists think T. rex had one of the most powerful bites of any land animal that

has ever lived, strong enough to crunch straight through solid bone. Its little arms look silly, it is true, but they were not weak. They were short and thick and could probably curl a heavy weight. Scientists still argue about what they were for.

And here is something that would surprise the people who made that roaring robot. Many of T. rex's close cousins were covered in feathers, fluffy ones, more like a giant terrifying chicken than a scaly lizard. Some scientists think baby T. rex may have been fluffy too. Nobody knows for certain, which is exactly what makes it fun to think about.

Now come and meet Sophie. Sophie is a Stegosaurus, one of those dinosaurs with the row of tall bony plates down its back and the spikes on its tail. And Sophie is special, because she is the most complete Stegosaurus skeleton ever found anywhere in the world. When scientists dug her up, more than four out of every five of her bones were still there, over three hundred of them, which almost never happens. Most dinosaur skeletons in museums are missing lots of pieces, filled in with copies, but with Sophie, most of what you see is the real thing.

She went on show here in twenty fourteen, and she was the first brand-new, nearly complete dinosaur to go on display in this museum in almost a hundred years. Nobody even knows if Sophie was a boy or a girl. Those plates on her back may have helped her show off to other stegosaurs, or soak up the sun's warmth, or scare off attackers. So the museum gives you both, a robot T. rex to make you jump, and Sophie, real bones, standing quietly, a hundred and fifty million years old.

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An Earthquake, a Giant Sloth, and Stardust

Three more wonders. First, in the volcanoes and earthquakes zone, stand in the earthquake room: a mock-up of a Japanese supermarket that shakes beneath your feet, replaying the real 1995 Kobe earthquake while the shelves rattle around you. Second, hunt for the giant ground sloth skeleton — a sloth the size of an elephant that lived in South America until just a few thousand years ago. Charles Darwin dug up bones of giant sloths on his famous voyage, and puzzling over creatures like these helped lead him to his theory of evolution. The museum owns thousands of specimens Darwin collected himself. Third, in the Minerals gallery and the Vault, find the meteorites — actual rocks from space, older than the Earth itself. Some contain tiny grains of stardust that formed around other stars before

our Sun even existed. You are allowed to say "whoa." And a building secret as you walk around: this whole museum is decorated like a stone zoo. The architect covered the walls, inside and out, with carved animals — monkeys climbing the arches, fish, birds, and creatures on every column. Living animals decorate the west side of the building; extinct ones decorate the east. See how many carved creatures you can spot. Nobody has ever counted them all.

Standing on a Shaking Floor

Walk into the little Japanese supermarket in the earthquakes gallery and put your hand on the rail, because in a moment the floor is going to move. What you are about to feel is a copy of a real disaster. On the seventeenth of January, nineteen ninety-five, a powerful earthquake struck the city of Kobe in Japan, early in the morning while people were still waking up. It measured around seven on the scale scientists use, which is very strong, and it shook the ground so hard that buildings, roads, and railways came crashing down.

Inside the mock supermarket, the shelves begin to rattle, tins and packets tremble, the lights flicker, and on a little screen you can watch real footage from that morning in Kobe. It only lasts a few seconds, but it is enough to understand something you cannot learn from a book, what it actually feels like when the solid, dependable ground turns wobbly beneath your feet.

So why does the Earth do this? Here is the secret. The ground you are standing on is not one solid ball of rock. The Earth's hard outer shell is cracked into enormous pieces, like a cracked eggshell, and these pieces, called plates, are floating very slowly on hotter, softer rock below. They are always moving, about as fast as your fingernails grow, far too slow to feel. But sometimes two plates get stuck against each other, pushing and pushing for years, and the pressure builds and builds. Then, suddenly, they slip. All that stored-up energy is let go at once, and the ground shakes. That is an earthquake.

The very same cracking and pushing also makes volcanoes, which is why this gallery keeps them together. Melted rock from deep inside the Earth forces its way up through gaps between the plates and bursts out as a volcano. Earthquakes and volcanoes are twins, both of them made by the restless, moving skin of our planet.

So the shaking floor is not just a fairground ride. It is a tiny, safe taste of one of the most powerful forces on Earth.

And when it stops, and the supermarket goes still again, you can remember that somewhere under your feet, right now, those giant plates of rock are still slowly grinding along, building up the pressure for the next one, somewhere, someday. The Earth is never really holding still.

A Feather, the Moon, and a Rock From the Sky

There is a room in this museum full of the rarest treasures of all, and three of them are so amazing they are worth hunting for. The first looks small and grey and easy to walk past, but it may be the most precious fossil in the whole building. It is called *Archaeopteryx*, and it is the fossil of a creature that is half dinosaur and half bird.

Look closely and you can see feathers pressed into the stone, actual feathers, from an animal that lived about a hundred and fifty million years ago. But this was no ordinary bird. It had sharp teeth in its mouth, claws on its wings, and a long bony tail like a lizard. It is one of the best clues we have ever found that birds are the living cousins of the dinosaurs, that the pigeon on the pavement outside is a tiny, feathery, far-off relative of the dinosaurs in the gallery. This particular fossil was found in Germany in eighteen sixty-one, and it is so famous that scientists simply call it the London specimen, because this is where it lives.

The second treasure is even further from home, because it is not from Earth at all. It is a piece of the Moon. When American astronauts flew to the Moon on the very last Apollo mission, they brought back rocks, and a tiny fragment was given to the United Kingdom as a gift. It is only about the size of your thumbnail, mounted on a little plaque beside a small British flag that actually flew all the way to the Moon and back. So you are looking at something that a human being picked up off the surface of another world.

And the third treasure fell out of the sky all by itself. In seventeen ninety-five, in a field in Yorkshire, a farmworker heard a whistling roar and saw a stone come crashing down out of the clouds, so close that it flung soil into the air. That stone is here. Back then, most people flatly refused to believe that rocks could fall from space at all. This one, called the *Wold Cottage meteorite*, helped scientists prove that they can, that lumps of rock really do drift through space and sometimes land on Earth.

So in a single room you can meet a feathered dinosaur, rest your eyes on a piece of the Moon, and stand before a rock that fell flaming from the sky. Three treasures, three impossible journeys, all of them ending right here.